

Fig. 3: An actual-size, single-side PCB for the wireless microphone

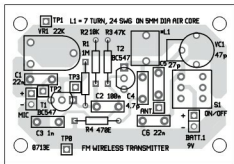


Fig. 4: Component layout for the PCB in Fig. 3

The output of the FM kit is connected to the audio amplifier made using IC1. To make IC1 work on a single power supply, a pair of diodes D1 and D2 in conjunction with 100-kilo-ohm resistor R8 is used to create a virtual earth. Heat-sink should be used to prevent shutdown of IC1 due to overheating.

Working of the circuit is simple. Once transmitter and receiver are tuned (see calibration section), the microphone picks up the teacher's voice and transmits it over the FM band. The FM kit receives the audio, which is further amplified by the amplifier. This amplified audio is available at loudspeaker LS1.

Calibration

Keep the transmitter and receiver sections around three metres apart.

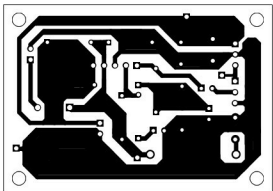


Fig. 5: An actual-size, single-side PCB for the receiver amplifier

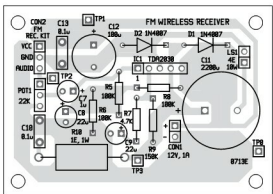


Fig. 6: Component layout for the PCB in Fig. 5

First, switch on the receiver and tune it to some fixed frequency that is away from the regular broadcasting channel frequencies. Set volume in the middle position.

Now turn on the wireless microphone with switch S1. With the help of a small plastic screwdriver, turn trimmer VC1 a little at a time (5 degrees maximum) until 'hissing' from the radio stops and null point is found. This means that the wireless microphone is now tuned at the same frequency.

Place a quartz clock very close to the electret microphone. Now you should be able to hear 'clicks' on loudspeaker LS1. The 'clicks' sound should

be very clear and without noise. If not, use preset VR1 to set the sensitivity and/or spread/compress the coil. Repeat this process two-three times to fine-tune both the circuits. Ensure that the aligned frequency does not interfere with any local FM station.

Construction and testing

An actual-size, single-side PCB for the wireless microphone is shown in Fig. 3 and its component layout in Fig. 4. PCB for the receiver amplifier is shown in Fig. 5 and its component layout in Fig. 6. Suitable connectors are provided on the PCB for input and output. Assemble the circuits on these PCBs only to minimise errors. Carefully assemble the components and double-check for any overlooked error.

For wireless microphone use a small plastic box that can fit inside your shirt's front pocket. For receiver use a cabinet suitable for standard speakers. Fix the receiver amplifier inside and connect a 30.5cm (12-inch) long telescopic antenna or an insulated 18SWG wire for antenna.

To test the wireless microphone for proper functioning, check correct power supply at TP1 with respect to TP0. Audio picked up by the microphone can be checked at TP2. The amplified audio can be checked at TP3 using an oscilloscope. Similarly, check the receiver amplifier side too for correct voltages and audio signals as per the test point table. ●

The author is a regular contributor to EFY

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